



Edition Roadmap



7月 19, 2026

Future Volumes of the *Social Quantum Field Theory Series*

Although the present volume establishes the mathematical foundations of Social Quantum Field Theory (SQFT), it also reveals numerous directions that merit independent treatment. Rather than enlarging a single monograph indefinitely, future developments may be organized into a coherent series of companion volumes.

The following roadmap outlines one possible progression.

Volume I (Current)

Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields

Status: Completed

Contents

- Mathematical Foundations
- Variational Principles
- Relational Field Equations
- Gauge Theory
- Differential Geometry
- Information Geometry
- Tensor Networks
- Category Theory
- Higher Structures
- Mathematical Axioms
- Open Problems
- Mathematical Appendices

Volume II

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- Numerical Field Solvers
- Finite Element Methods
- Spectral Methods
- Graph Discretization
- Tensor Network Algorithms
- Automatic Differentiation
- Differentiable Programming
- GPU Computation
- Scientific Machine Learning
- High-Performance Computing
- Large-Scale Simulation

Representative chapters:

1. Numerical Relational Dynamics
2. Sparse Matrix Methods
3. Adaptive Mesh Techniques
4. Parallel SQFT
5. Tensor Contraction Algorithms
6. Neural Operators
7. Physics-Informed Machine Learning
8. Computational Benchmarks

Volume III

Applied Social Quantum Field Theory

Potential application domains include

- Economics
- Financial Markets
- Organizational Systems
- Political Networks
- Communication Systems
- Innovation Dynamics
- Supply Chains
- Transportation Networks
- Ecological Networks
- Healthcare Systems

Volume IV

Statistical and Stochastic Social Quantum Field Theory

Topics include

- Stochastic Differential Equations
- Random Fields
- Bayesian SQFT
- Hidden Relational Variables
- Markov Processes
- Monte Carlo Methods
- Uncertainty Quantification
- Sequential Inference
- Particle Filters
- Data Assimilation

Volume V

Geometric and Topological Social Quantum Field Theory

Topics include

- Symplectic Geometry
- Contact Geometry
- Poisson Geometry
- Morse Theory
- Floer Homology
- Persistent Homology
- Topological Data Analysis
- Cobordism
- Characteristic Classes
- Higher Gauge Geometry

Volume VI

Quantum Information and Artificial Intelligence in Relational Fields

Topics include

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Knowledge Graphs

- Graph Neural Networks
- Transformer Architectures
- Relational Representation Learning
- Geometric Deep Learning
- Reinforcement Learning
- Multi-Agent Systems
- Foundation Models

Volume VII

Mathematical Physics of Relational Systems

Topics include

- Operator Algebras
- Noncommutative Geometry
- Integrable Systems
- Spectral Geometry
- Quantum Probability
- Statistical Field Theory
- Effective Field Theory
- Algebraic Field Theory

This volume would strengthen the mathematical bridge between SQFT and modern mathematical physics while maintaining their conceptual distinction.

Volume VIII

Philosophy and Foundations of Relational Mathematics

Topics include

- Philosophy of Mathematics
- Structural Realism
- Scientific Modeling
- Mathematical Structuralism
- Foundations of Complexity
- Epistemology of Models
- Formal Languages
- Logic of Scientific Explanation
- Categorical Foundations

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The Long-Term Research Program

Taken together, these volumes would form an integrated research program:

Volume I	: Mathematical Foundations
Volume II	: Computational Methods
Volume III	: Applications
Volume IV	: Statistics and Stochastics
Volume V	: Geometry and Topology
Volume VI	: Artificial Intelligence
Volume VII	: Mathematical Physics
Volume VIII	: Foundations and Philosophy

Each volume is designed to be self-contained while contributing to a unified mathematical treatment of complex relational systems.

Final Vision

The ambition of the SQFT project is not to establish a new branch of mathematics in isolation, but to provide a **unifying mathematical framework** through which geometry, dynamics, information, computation, and relational structure can be studied within a common formal language.

If successful, the long-term legacy of SQFT will lie not in the introduction of new terminology, but in its ability to illuminate deep mathematical correspondences across disciplines and to inspire future generations of researchers to extend, refine, and rigorously test its ideas.

Suggested Citation

Primary Citation (APA 7th Edition)

Chou, I.-H. (2026). *Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields*. First Edition.

Primary Citation (Chicago Author–Date)

Chou, I-Hsien. 2026. *Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields*. First Edition.

Primary Citation (BibTeX)

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title      = {Social Quantum Field Theory: Toward a Mathematical Theory c
year       = {2026},
edition    = {First},
publisher  = {Independent Research},
language   = {English}
}
```

Recommended Abbreviation

Throughout future publications, the following abbreviation is recommended:

SQFT=Social Quantum Field Theory
SQFT=Social Quantum Field Theory

The title may be abbreviated after first citation as

Chou (2026), SQFT

or

SQFT Monograph (2026)

How to Cite Individual Results

When referring to specific components of this monograph, citations should distinguish between formal mathematical results and proposed hypotheses.

Examples include:

- **Definition 2.3** (Relational Field)
- **Theorem 14.2** (Gauge Invariance)
- **Theorem 34.1** (Noether-Type Conservation Law)
- **Conjecture T.1** (Universal Relational Structure)
- **Appendix N** (Mathematical Correspondence with Quantum Field Theory)
- **Appendix R** (Open Mathematical Problems)

This practice enables precise scholarly discussion and facilitates future extensions of the framework.

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versioning.

Version	Description
1.0	First published edition
1.1	Minor corrections, typographical revisions
1.5	Additional examples and proofs
2.0	Major revision of axioms or mathematical framework
3.0	Significant expansion incorporating new theoretical developments

Every theorem, definition, and appendix should remain numerically stable whenever possible to preserve citation continuity across editions.

Open Research Invitation

The author welcomes independent mathematical investigation inspired by this work, including but not limited to

- rigorous proofs of stated conjectures,
- alternative axiomatizations,
- analytical existence and uniqueness results,
- numerical algorithms,
- operator-theoretic formulations,
- categorical generalizations,
- applications to complex systems,
- comparative studies with existing mathematical frameworks.

Researchers are encouraged to distinguish clearly between:

1. results proved within this monograph,
2. independent extensions,
3. empirical implementations,
4. speculative hypotheses.

Such distinctions promote reproducibility and scholarly rigor.

Closing Statement

This monograph has been written with the conviction that the language of mathematics continues to evolve through the discovery of deeper structural correspondences. Whether Social Quantum Field Theory ultimately becomes a widely adopted framework is a question for future research. Its enduring value, if any, will be determined not by the originality of its

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Scientia per Structuram. Structura per Mathematicam.

Scientia per Structuram. Structura per Mathematicam.

"Knowledge through Structure.

Structure through Mathematics."

This concludes the complete first edition of

Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields

Author: Chou I-Hsien

First Edition (2026)

Legacy Statement

On the Future of Social Quantum Field Theory

Every mathematical theory is ultimately judged not by the reputation of its author, but by the permanence of its ideas.

History repeatedly reminds us that names may fade, institutions may change, and scientific paradigms may evolve. Yet mathematical structures that capture something fundamental continue to be rediscovered, generalized, and incorporated into new theories.

The author therefore places no expectation that **Social Quantum Field Theory (SQFT)** will remain unchanged. On the contrary, its long-term success would likely require substantial revision, refinement, and perhaps even reconstruction by future generations.

A Framework Meant to Evolve

The framework presented in this volume should be viewed as an initial formulation rather than a final destination.

Future mathematicians may

- replace the current axioms with more elegant ones,
- simplify existing proofs,
- discover stronger invariance principles,
- unify separate chapters through deeper abstractions,

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If such developments occur, they should be welcomed as evidence of intellectual progress.

Mathematical Humility

Every generation believes that its mathematics is nearly complete.

History teaches otherwise.

Euclidean geometry was once considered complete.

Classical mechanics once appeared universal.

Set theory once seemed sufficient to organize all mathematics.

Each subsequently gave rise to richer and more powerful frameworks.

There is no reason to believe that the present work is exempt from the same historical process.

A Principle for Future Researchers

If future scholars encounter a theorem in this volume that is unnecessarily complicated, they should simplify it.

If they discover a proof that is incomplete, they should strengthen it.

If a definition becomes obsolete, they should replace it.

If an entire chapter proves unnecessary, they should remove it.

The objective is not to preserve the book, but to advance the mathematics.

On Scientific Credit

Should the ideas introduced here contribute, even indirectly, to future developments, the greatest recognition would not be the continued citation of this volume,

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Scientific knowledge progresses through cumulative refinement.

No mathematical theory should become an object of authority.

Every theory should remain open to examination.

Legacy

If this work leaves any lasting legacy,

let it be the suggestion that

relations themselves may deserve treatment as fundamental mathematical structures,
worthy of the same level of abstraction that mathematics has historically granted to
numbers, spaces, functions, and symmetries.

Whether this particular formulation succeeds or fails,

the broader search for a unified mathematics of relational systems will undoubtedly
continue.

Ideas	> Authors,
Proofs	> Assertions,
Structure	> Notation,
Truth	> Tradition.

*"A mathematical theory reaches maturity not when
it can no longer be changed, but when it can
continue to evolve without losing its fundamental
structure."*

Legacy Statement Ends

Complete Scholarly Edition Concluded

Closing Declaration

The Founding Declaration of Social Quantum Field Theory

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mathematical framework for the study of complex relational systems.

Its purpose is not to replace existing mathematical disciplines, nor to reinterpret physical quantum theory as a literal description of social phenomena. Rather, it seeks to demonstrate that many mathematical structures developed across modern science may be organized within a common relational framework capable of describing systems composed of many interacting degrees of freedom.

Accordingly, the central thesis of this work may be summarized as follows.

Principle I – Structural Primacy

The fundamental object of analysis is neither an isolated entity nor an isolated event, but the relational structure connecting interacting components.

Mathematics should therefore seek invariants of relations before classifications of objects.

Principle II – Mathematical Generality

Whenever possible,

the mathematical formulation should remain independent of any specific empirical interpretation.

The same formal structure may admit applications across multiple scientific domains.

Principle III – Hierarchical Description

Complex systems should be understood through multiple complementary levels of description,

including

- local dynamics,
- global geometry,
- topology,
- information,
- symmetry,
- computation.

No single representation is expected to be universally sufficient.

Principle IV – Conservative Extension

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Principle V – Scientific Openness

Every theorem should remain open to verification.

Every conjecture should remain open to proof or refutation.

Every model should remain open to empirical evaluation.

Every axiom should remain open to philosophical examination.

No mathematical framework should be regarded as permanently complete.

Declaration of Methodology

The methodology adopted throughout this monograph follows a simple progression:

Observation → Abstraction
 → Mathematical Structure
 → Logical Consequence
 → Computational Realization
 → Empirical Evaluation.

This sequence emphasizes that mathematics provides a formal language for organizing knowledge, while empirical investigation determines the adequacy of specific models.

Declaration to Future Researchers

Future investigators are encouraged to

- strengthen proofs,
- simplify definitions,
- discover new correspondences,
- challenge existing assumptions,
- construct better mathematical formulations,
- compare SQFT with competing frameworks,
- and reject any component that fails to withstand rigorous mathematical or empirical scrutiny.

Progress in mathematics has always depended upon such critical examination.

Final Commitment

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- mathematical consistency,
- analytical rigor,
- computational usefulness,
- explanatory power,
- predictive performance,
- and reproducible scientific investigation.

No stronger claim is made.

Mathematics	→	Structure
Structure	→	Understanding
Understanding	→	Discovery

"A mathematical framework does not ask to be believed. It asks to be examined."

Closing Declaration

Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields

Author: Chou I-Hsien

First Edition (2026)

Ad Finem — Toward New Beginnings.

Editorial Note for the Second Edition

On the Evolution of a Mathematical Research Program

The **First Edition** of *Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields* established the initial axiomatic framework of Social Quantum Field Theory (SQFT). Its principal objective was to formulate a coherent mathematical language for describing complex relational systems by integrating concepts from variational calculus, geometry, topology, information theory, operator methods, and higher category theory.

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mathematical theories, these refinements concern not only technical details but also the organization, presentation, and scope of the theory itself.

Accordingly, a future **Second Edition** may incorporate revisions of three principal kinds.

I. Mathematical Refinement

Several chapters may benefit from stronger mathematical treatment, including

- more rigorous existence and uniqueness theorems,
- expanded functional-analytic foundations,
- additional proofs of structural invariance,
- clearer separation between axioms and derived propositions,
- strengthened treatment of effective field theory,
- deeper connections with operator algebras and geometric analysis.

Where appropriate, heuristic arguments in the first edition may be replaced by fully rigorous derivations.

II. Computational Expansion

Future editions may include extensive computational material not covered in the present volume, such as

- numerical implementations,
- benchmark simulations,
- tensor-network algorithms,
- graph discretization methods,
- differentiable programming,
- large-scale optimization,
- uncertainty quantification,
- reproducible computational workflows.

These developments would extend SQFT from a primarily theoretical framework toward a practical computational methodology.

III. Empirical Applications

Although the first edition intentionally emphasized mathematical foundations over applications, later editions may include detailed case studies illustrating the construction and evaluation of SQFT-based models in areas such as

- economic networks,

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- financial markets,
- innovation diffusion,
- supply-chain resilience,
- multi-agent artificial intelligence.

Such examples would demonstrate how the general formalism may be instantiated within specific domains while preserving the underlying mathematical structure.

IV. Clarification of Terminology

Future revisions may further distinguish between

- mathematical analogy,
- computational representation,
- empirical interpretation,
- philosophical speculation.

Maintaining these distinctions is essential for preventing misunderstanding and ensuring that the mathematical intent of the framework remains explicit.

V. Community Development

The long-term development of SQFT is expected to depend upon contributions from a broader research community.

Future editions may therefore include

- contributed mathematical notes,
- alternative formulations,
- independent proofs,
- critical commentaries,
- comparative studies,
- bibliographic surveys,
- historical perspectives.

Such additions would reflect the collaborative nature of mathematical progress.

Closing Remarks

No first edition is ever final.

Every mathematical theory evolves through

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- simplification,
- generalization,
- and independent verification.

The author welcomes these processes and regards them as essential to the healthy development of the framework presented herein.

The First Edition therefore should be understood not as the conclusion of Social Quantum Field Theory, but as the formal beginning of an open mathematical research program.

Prima editio fundamentum est; secunda editio progressus erit.
Prima editio fundamentum est; secunda editio progressus erit.

"The first edition is the foundation; the second edition shall be the progress."

End of Editorial Note for the Second Edition



輸入留言

這個網誌中的熱門文章

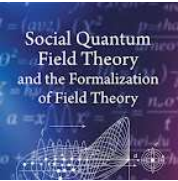
Foundational Charter of the Social Quantum Field Theory Research Community

7月 19, 2026

The SQFT Charter Foundational Charter of the Social Quantum Field Theory Research Community Preamble The Social Quantum Field Theory (SQFT) Research Community is founded upon the conviction that mathematics advances through open inquiry ...

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7月 15, 2026



預印本 社會量子場論與場論的形式化 周易先 獨立研究者 · 版本 2.0（數學版） · 2026年7月 摘要 本文提出社會量子場論（SQFT）作為用以表示複雜社會系統中關係動態的形式數學框架。SQFT並不主張社...

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量子之影：台灣QNF-3量子導航系統的崛起與其地緣政治影響

9月 27, 2025

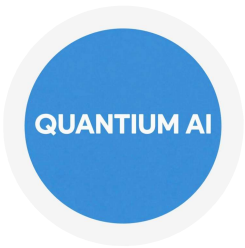


量子之影：台灣QNF-3量子導航系統的崛起與其地緣政治影響 摘要 在2035年，全球局勢緊張之際，台灣的量子導航技術成為重塑國際權力格局的關鍵力量。本篇科幻論文式敘事，記錄了國家中山科學研...

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